National College of Ireland

H9AIDM: AI Driven Decision Making

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Creating an Index Fund Using AI-Driven Decision-Making Approaches: Team Contribution Matrix

Project Overview

This document outlines the distribution of responsibilities and contributions for our index tracking portfolio optimization project, showcasing how team members collaborated to develop multiple approaches to solving this financial challenge.

Project Timeline

Project Phase	Timeframe	Key Milestones
Planning & Research	Week 1(Mar 3 - Mar 9, 2025)	Project scope finalized Literature review completed Research methodologies selected
Data Collection & Preprocessing	Week 2(Mar 10 - Mar 16, 2025)	Data sources identified Raw data collected Preprocessing pipeline completed
Algorithm Development	Weeks 3-4(Mar 17 - Mar 30, 2025)	Optimization model implemented Clustering approach developed Initial testing completed
Testing & Refinement	Week 5(Mar 31 - Apr 6, 2025)	Parameter tuning completed Performance benchmarking Approach integration
Analysis & Documentation	Week 6(Apr 7 - Apr 13, 2025)	Results analysed Report drafted Visualizations created
Final Delivery	Week 7(Apr 14, 2025)	Final report submission Presentation delivered Code repository finalized

Individual Contributions

Area of Work	Shoban Ravichandran	Mehwish Khatib	Collaborative Work
Project Planning	• Led initial project scoping	• Developed evaluation metrics	• Defined project scope and objectives

	• Defined research questions Established project timeline	• Created workflow architecture • Set up version control system	• Established key performance indicators • Aligned on success criteria
Research & Analysis	• Conducted literature review on index tracking methods • Researched optimization constraints • Analysed existing optimization models	• Researched clustering applications in finance • Identified relevant stock features for clustering • Investigated performance benchmarks	• Identified gaps in existing approaches • Formulated hybrid methodology • Selected appropriate benchmark indices
Data Management	• Defined data requirements • Data validation and cleaning • Feature selection for optimization	• Led data collection pipeline • Built data preprocessing workflow • Created feature engineering process • Managed data storage architecture	• Regular data quality reviews • Established shared data access protocols • Designed data documentation standards
Technical Implementation	• Developed AMPL optimization model • Implemented constraint formulations • Created objective function variations • Tuned optimization parameters • Built performance evaluation framework	• Implemented clustering-based approach • Developed K-means algorithm for stock selection • Created feature weighting mechanisms • Built visualization tools	• Integrated approaches for comparison • Troubleshooted implementation challenges • Code reviews and quality assurance • Performance benchmarking

	• Implemented comparative analysis framework		
Testing & Evaluation	• Tested optimization model robustness • Conducted sensitivity analysis • Analysed tracking error metrics	• Evaluated clustering algorithm performance • Tested different q values across approaches • Created performance visualizations	• Designed experimental framework • Compared results across methodologies • Identified strengths and limitations
Documentation & Reporting	• Authored report sections: Introduction, Problem Definition, Optimization Methodology. • Contributed to Results & Conclusion	• Authored report sections: Clustering Methodology, Evaluation Framework, Created data visualizations. • Contributed to Results & Conclusion	• Integrated final report • Created presentation materials • Cross-reviewed all sections • Prepared executive summary

Project Management Approach

The team employed an agile approach with weekly monitoring and regular check-ins to ensure alignment and progress. Git version control facilitated parallel development, with each team member maintaining responsibility for specific components while reviewing others' contributions.

Weekly Time Allocation (Per Team Member)

Activity	**Hours Per Week**	**Total Project Hours**
Research & Planning	10-12 hours (Week 1)	~12 hours
Coding & Implementation	10-15 hours (Weeks 2-4)	~25 hours
Testing & Analysis	12-15 hours (Weeks 4-5)	~20 hours
Documentation & Reporting	10-15 hours (Weeks 6-7)	~25 hours
Team Meetings & Collaboration	1-2 hours (throughout)	~12 hours
Total Time Investment	~20 hours/week average	~ (90 – 100) hours per person

Key Innovations & Contributions

- **Shoban Ravichandran**: Developed a novel constraint formulation in the optimization model that reduced tracking error by 15% compared to baseline approaches
- **Mehwish Khatib**: Created an innovative feature weighting mechanism for the clustering algorithm that improved stock selection efficiency
- **Joint Innovation**: Successfully integrated optimization and clustering approaches to create a hybrid methodology that demonstrated superior performance in volatile market conditions

Skills Leveraged

- **Shoban Ravichandran**: Mathematical optimization, financial modeling, performance analysis, technical writing
- **Mehwish Khatib**: Machine learning, data preprocessing, visualization, algorithm implementation
- **Shared Skills**: Problem-solving, financial analysis, programming, research methodology